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SARDAR PATEL UNIVERSITY

US04CBCA25 – Data Structure - II

External Examination (April-2026)

SYBCA (Sem-4)



Date: 04.04.2026

Time: 02.00 PM - 05.00 PM

Marks: 70

Day: Saturday

[10]

Q-1 MCQ

1. Leaf node is known as _____ node.
A. Terminal
B. Parent
C. Root
D. Non-terminal
2. A binary tree has at most _____ child.
A. 1
B. 2
C. 3
D. None of these
3. A linked list is _____ type of data structure.
A. Linear
B. Non-Linear
C. Both (A) and (B)
D. None of these
4. A linked list in which last node pointing to the first node is known as ...
A. Singly linked list
B. Circular linked list
C. Doubly linked list
D. None of these
5. _____ is the process to arrange the data in ascending or descending order.
A. Sorting
B. Searching
C. Inserting
D. Updating
6. Quick sort is also known as...
A. merge sort
B. heap sort
C. bubble sort
D. None of these
7. The _____ is an area into which data records are written when the file is first created.
A. Control area
B. Index area
C. Prime area
D. Overflow area
8. The open addressing and chaining are two classes of _____ techniques.
A. Collision Resolution
B. Hashing
C. Blocking
D. None of these
9. The number of sub tree of a node in a tree is called _____ of a node.
A. Degree
B. Level
C. Height
D. Index
10. Which design technique is used for Quick Sort?
A. Divide and Conqueror
B. Dynamic Programming
C. Greedy
D. Backtracking

Q-2 Give answer in short. (ANY TEN)

1. Define Traversals. List the types of binary tree traversal.
2. Define tree terms with an example: Tree, Root Node, Leaf Node, and Level.
3. What is Binary Tree? Also give its example.
4. What do you mean by singly linked list?
5. Explain in brief: Doubly linked list.
6. What is a circular linked list?
7. List out application of sorting.
8. Define searching. List out the names of searching techniques.
9. Define sorting. List out the names of sorting techniques.
10. Define: Hashing function, Collision resolution technique.
11. Explain in brief prime area.
12. List types of Chaining.

- Q-3** A. Write down an algorithm for preorder traversal of a binary tree. [5]
 B. Explain the insertion of a node in a binary with an example. [5]

OR

- Q-3** A. Write down an algorithm for inorder traversal of a binary tree. [5]
 B. Discuss the internal memory representation of a binary tree using sequential representation. [5]

- Q-4** A. Write an algorithm to insert node at front position in singly linked list. [5]
 B. Write an algorithm to insert node at end position in singly linked list. [5]

OR

- Q-4** A. Write an algorithm to delete node at beginning in singly linked list. [5]
 B. Write an algorithm to delete node from end position in singly linked list. [5]
Q-5 A. Write down the algorithm of bubble sort. [5]
 B. Write down the algorithm of Linear Search. [5]

OR

- Q-5** A. Write down the algorithm of Quick sort. [5]
 B. Write down the algorithm of Binary Search. [5]
Q-6 A. Write a detail note on structure and processing of index sequential file supported by IBM. [10]

OR

- Q-6** A. List and explain popular hashing function with example. [10]

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